

Security at the Network Layer: IPSec

Introduction

- Security at the application layer and transport layer may not be enough,
- All client/server programs are not protected at the application layer
 - for example, PGP and S/MIME protect only electronic mail
- Not all client/server programs at the application layer use the service of TCP to be protected by SSL or TLS
 - some programs use the service of UDP

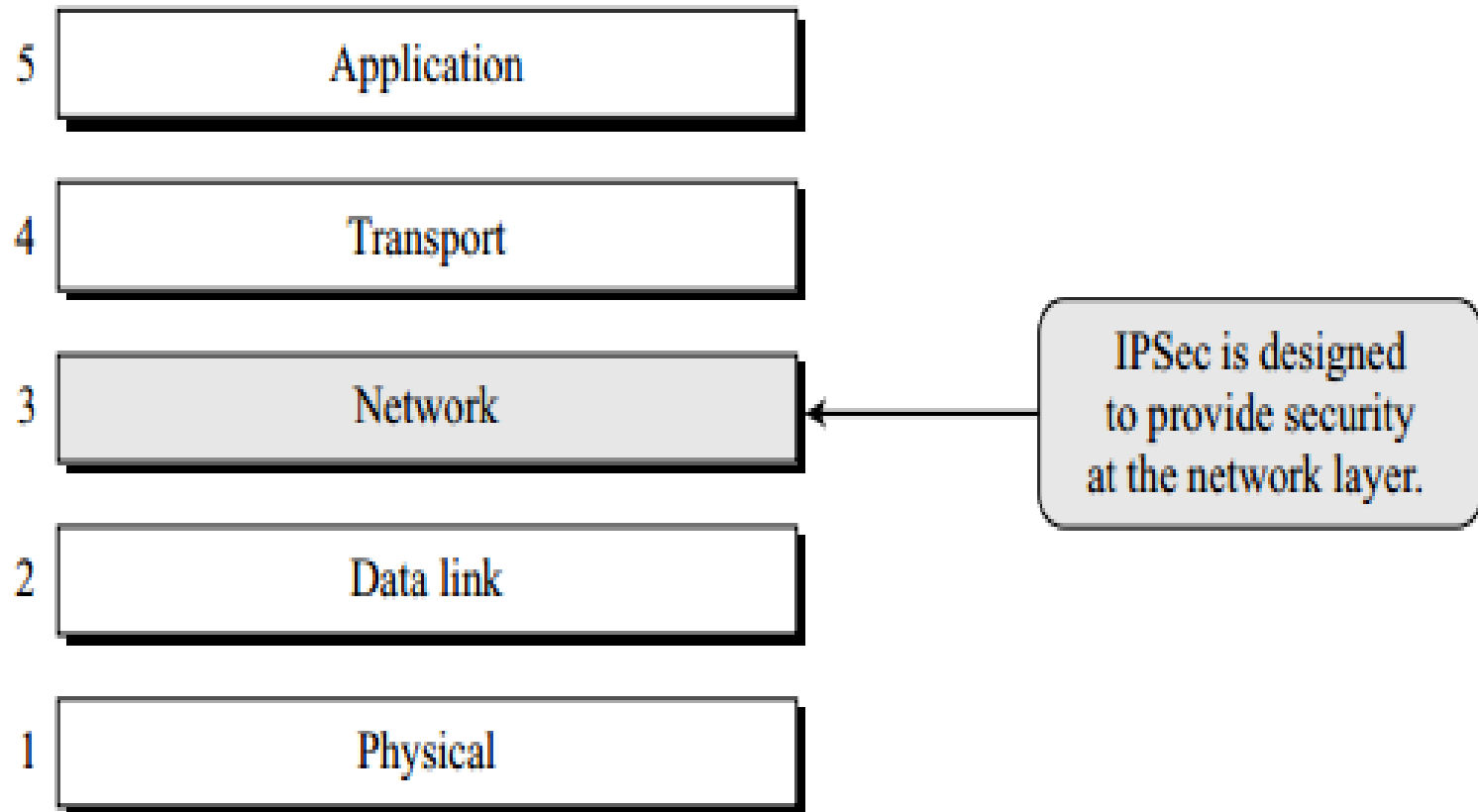
Introduction

- Many applications, such as routing protocols, directly use the service of IP
 - Security services at the IP layer is essential

Security at the Network Layer: IPSec

- IP Security (IPSec) is a collection of protocols designed by IETF to provide security for a packet at the network level.
- network layer in the Internet is often referred to as the Internet Protocol or IP layer.
- IPSec helps create authenticated and confidential packets for the IP layer

TCP/IP protocol suite and IPSec



Usage of IPSec

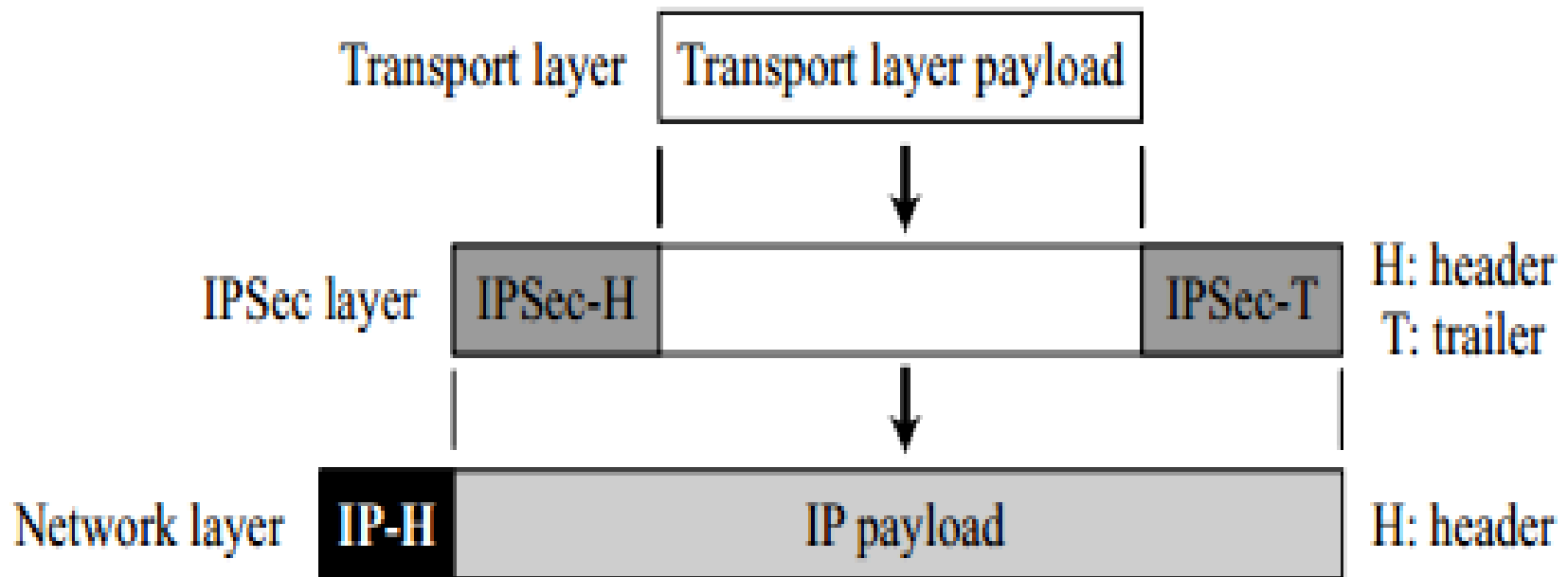
- It can enhance the security of those client/server programs who use their own security protocols (**electronic mail**)
- It can provide security for those client/server programs that do not use the security services provided at the transport layer (**http**).
- It can provide security for node-to-node communication programs (**routing protocols**)

Modes of IPSec

- IPSec operates in one of two different modes:
 - transport mode
 - tunnel mode

Transport Mode

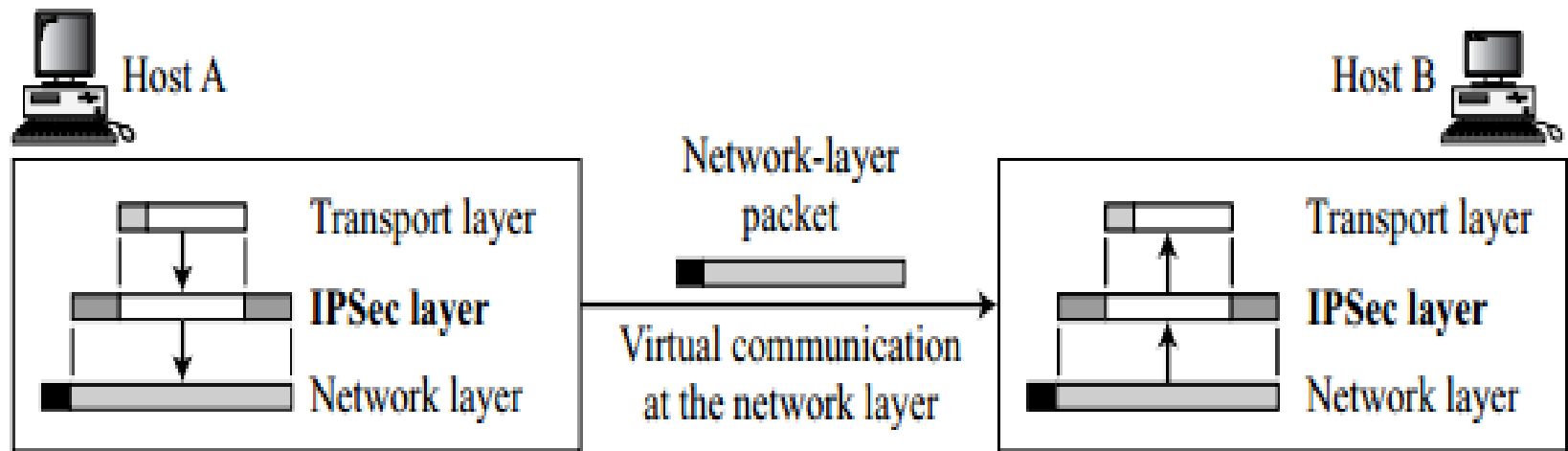
- The transport mode protects the network layer payload



Transport Mode

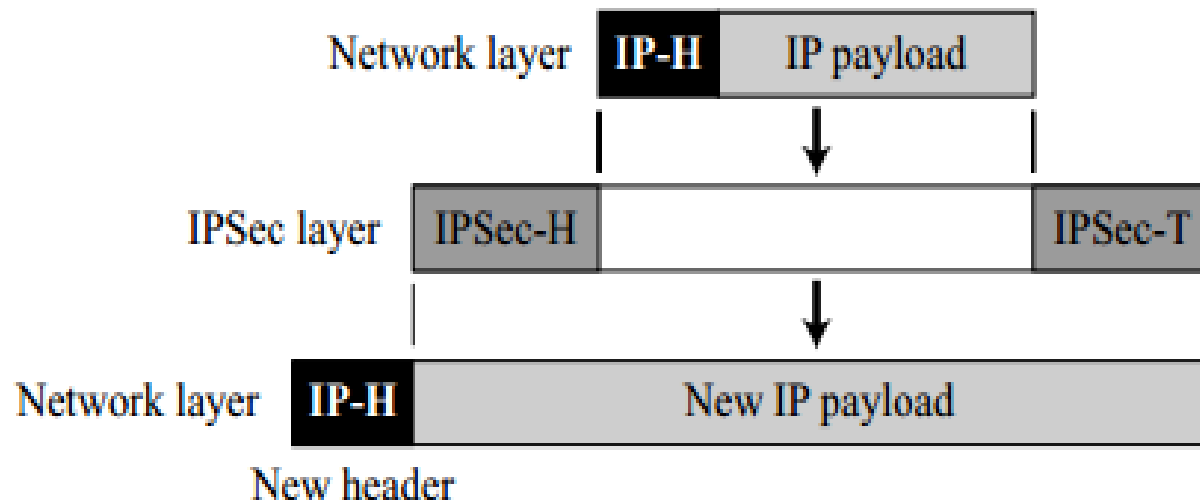
- Transport mode is normally used when we need host-to-host (end-to-end) protection of data
- IPSec in transport mode does not protect the IP header
 - it only protects the information coming from the transport layer (payload)

Transport mode operation



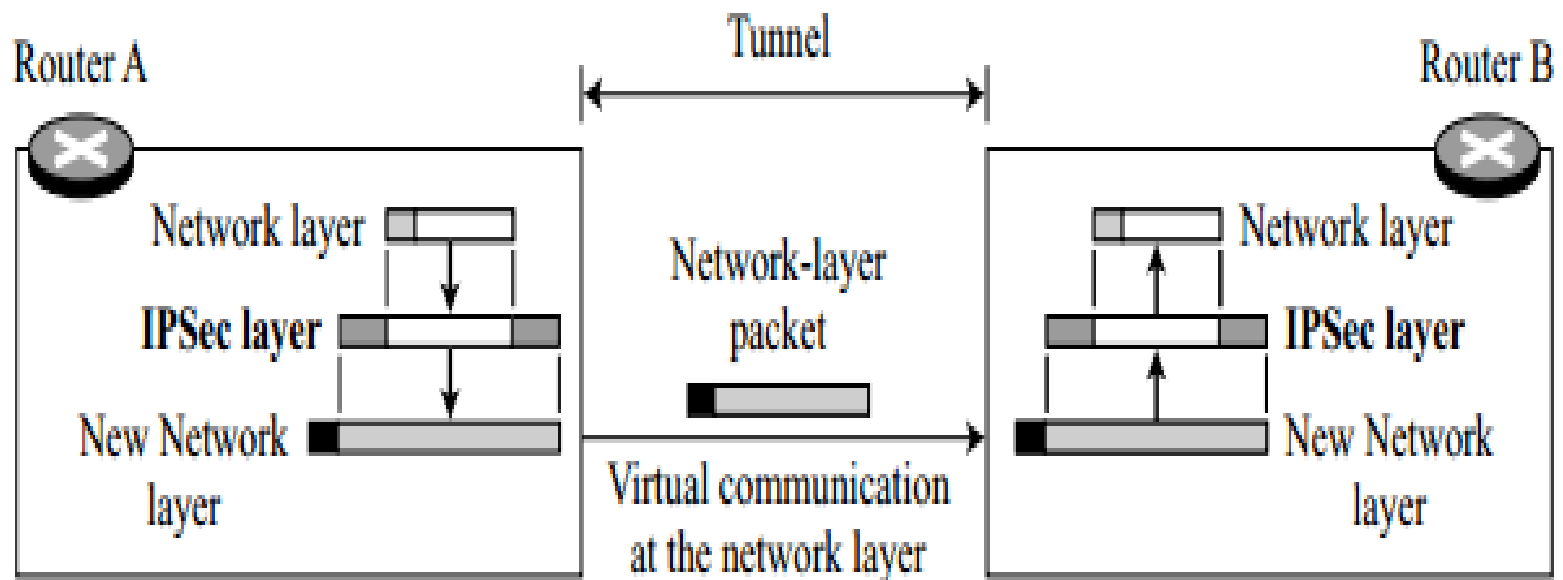
Tunnel Mode

- In tunnel mode, IPSec protects the entire IP packet (between the sender and the receiver)

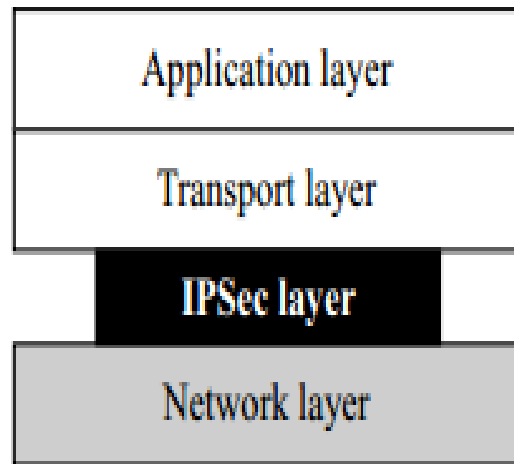


Tunnel Mode

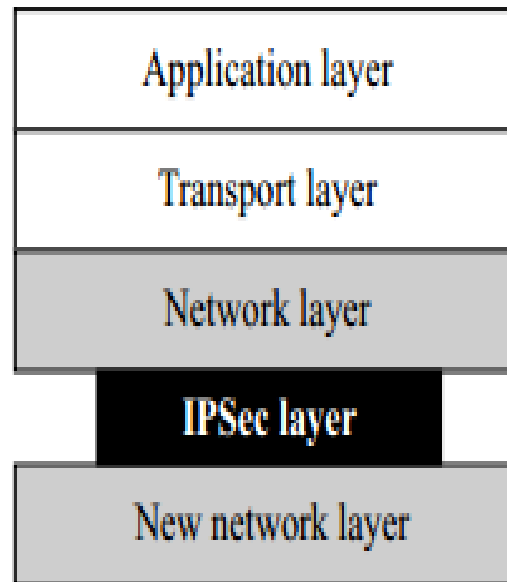
- Tunnel mode is normally used between two routers, between a host and a router, or between a router and a host



Transport mode versus tunnel mode



Transport Mode



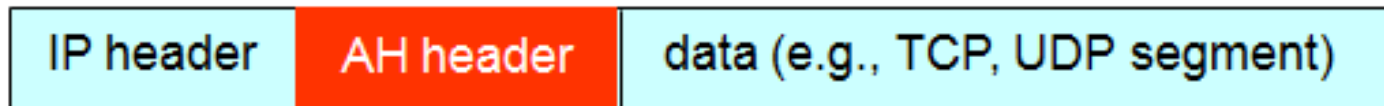
Tunnel Mode

Protocols of IPSec

- To provide authentication and/or encryption for packets at the IP level, IPSec defines two protocols
 - Authentication Header (AH) Protocol
 - Encapsulating Security Payload (ESP) Protocol

Authentication Header (AH) Protocol

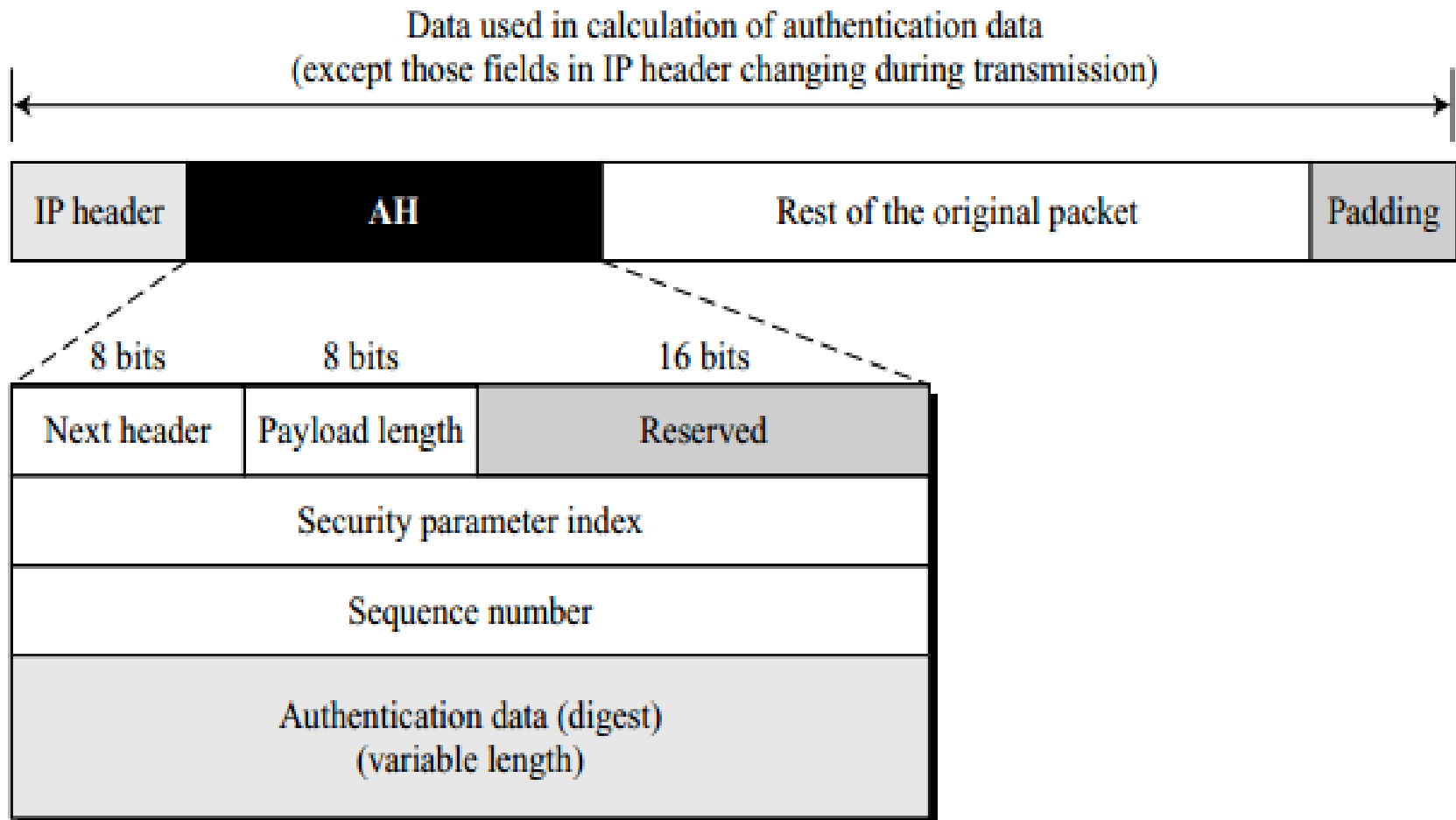
- Provides source authentication, data integrity, **no confidentiality**
- AH header inserted between IP header, data field.
- IP header protocol field: **51**
- Intermediate routers process datagrams as usual



Authentication Header (AH) protocol

- The protocol uses a hash function and a symmetric key to create a message digest
- The digest is inserted in the authentication header
- The AH is then placed in the appropriate location, based on the mode of IPSec

Authentication Header (AH)



Addition of Authentication Header

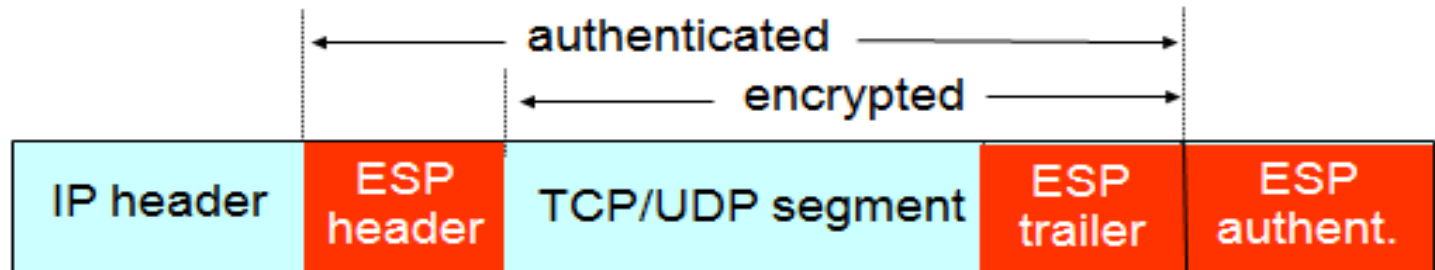
- An **AH** is added to the payload with the authentication data field set to 0.
- Padding may be added.
- Only those fields of the IP header that do not change during transmission are included in the calculation of the message digest (**authentication data**).
- The authentication data are inserted in the authentication header.
- The IP header is added after changing the value of the protocol field to 51.

AH header includes:

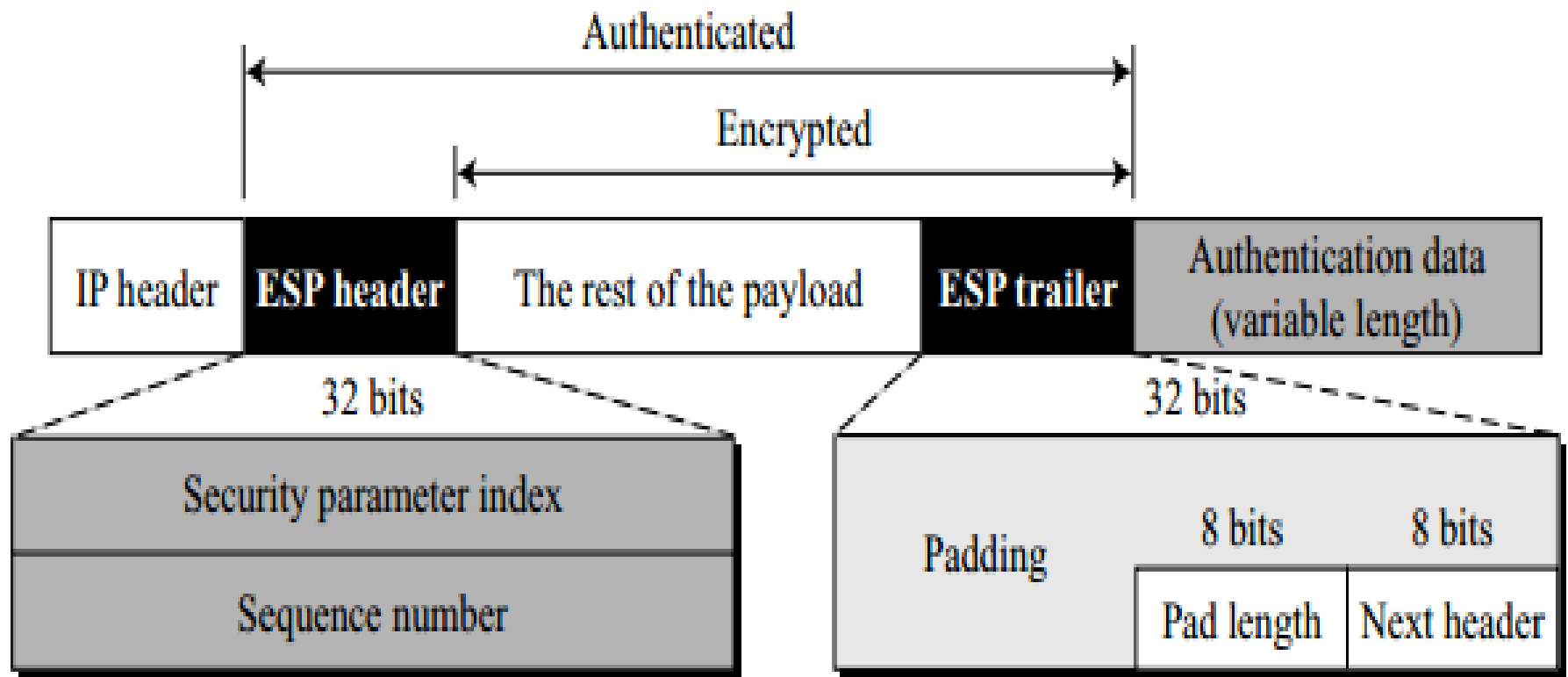
- **Next header** field specifies type of data (e.g., TCP, UDP, ICMP)
- **Payload length** field specifies the length of the authentication header
- **Security parameter index** (SPI) helps as a virtual circuit identifier
- **Authentication data** field is the result of applying a hash function to the entire IP datagram except for the fields that are changed during transit

Encapsulating Security Payload (ESP)

- Provides secrecy, host authentication, data integrity.
- Data, ESP trailer encrypted.
- Next header field is in ESP trailer.



Encapsulating Security Payload (ESP)



ESP procedure

- An ESP trailer is added to the payload.
- The payload and the trailer are encrypted.
- The ESP header is added.
- The ESP header, payload, and ESP trailer are used to create the authentication data.
- The authentication data are added to the end of the ESP trailer.
- The IP header is added after changing the protocol value to 50.

Encapsulating Security Payload (ESP)

- When an IP datagram carries an ESP header and trailer, the value of the protocol field in the IP header is **50**
- The next-header field of ESP trailer holds the original value of the protocol field

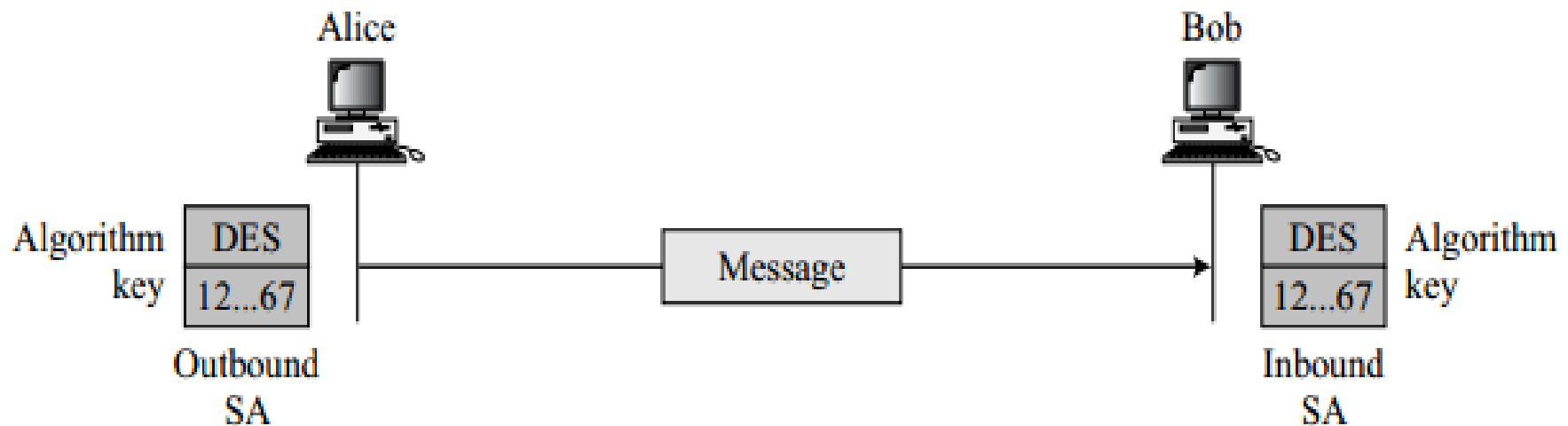
Services Provided by IPSec protocols

<i>Services</i>	<i>AH</i>	<i>ESP</i>
Access control	yes	yes
Message authentication (message integrity)	yes	yes
Entity authentication (data source authentication)	yes	yes
Confidentiality	no	yes
Replay attack protection	yes	yes

Security Association

- IPSec requires a logical relationship between two hosts, called a **Security Association (SA)**
- A Security Association is a contract between two parties
 - creates a secure channel between them

- A Simple **Security Association (SA)**



Security Association

- A Security Association(SA) can be very complex.
- To allow bidirectional communication, each site needs to have both inbound and outbound Sas
- **Solution**
 - set of SAs can be collected into a database, **Security Association Database (SAD)**

Security Association Database (SAD)

- The database is a two-dimensional table with each row defining a single SA
- There are two SADs, one inbound and one outbound.

Security Association Database (SAD)

< SPI, DA, P >							
< SPI, DA, P >							
< SPI, DA, P >							
< SPI, DA, P >							

Security Association Database

Legend:

SPI: Security Parameter Index

DA: Destination Address

AH/ESP: Information for either one

P: Protocol

Mode: IPsec Mode Flag

SN: Sequence Number

OF: Overflow Flag

ARW: Anti-Replay Window

LT: Lifetime

MTU: Path MTU (Maximum Transfer Unit)

- Each entry in an inbound SAD is selected using a triple index:
 - security parameter index
 - defines the SA at the destination
 - destination address
 - destination address of the host
 - protocol
 - IPSec requires that a destination define a different SA for each protocol (AH or ESP)

Typical SA Parameters

<i>Parameters</i>	<i>Descriptions</i>
Sequence Number Counter	This is a 32-bit value that is used to generate sequence numbers for the AH or ESP header.
Sequence Number Overflow	This is a flag that defines a station's options in the event of a sequence number overflow.
Anti-Replay Window	This detects an inbound replayed AH or ESP packet.
AH Information	This section contains information for the AH protocol: 1. Authentication algorithm 2. Keys 3. Key lifetime 4. Other related parameters
ESP Information	This section contains information for the ESP protocol: 1. Encryption algorithm 2. Authentication algorithm 3. Keys 4. Key lifetime 5. Initiator vectors 6. Other related parameters
SA Lifetime	This defines the lifetime for the SA.
IPSec Mode	This defines the mode, transport or tunnel.
Path MTU	This defines the path MTU (fragmentation).

Security Policy

- The Security Policy (SP) defines the type of security applied to a packet when it is to be sent or when it has arrived.
- Each host that is using the IPSec protocol needs to keep a **Security Policy Database (SPD)**
- There is a need for an inbound SPD and an outbound SPD

Security Policy Database (SPD)

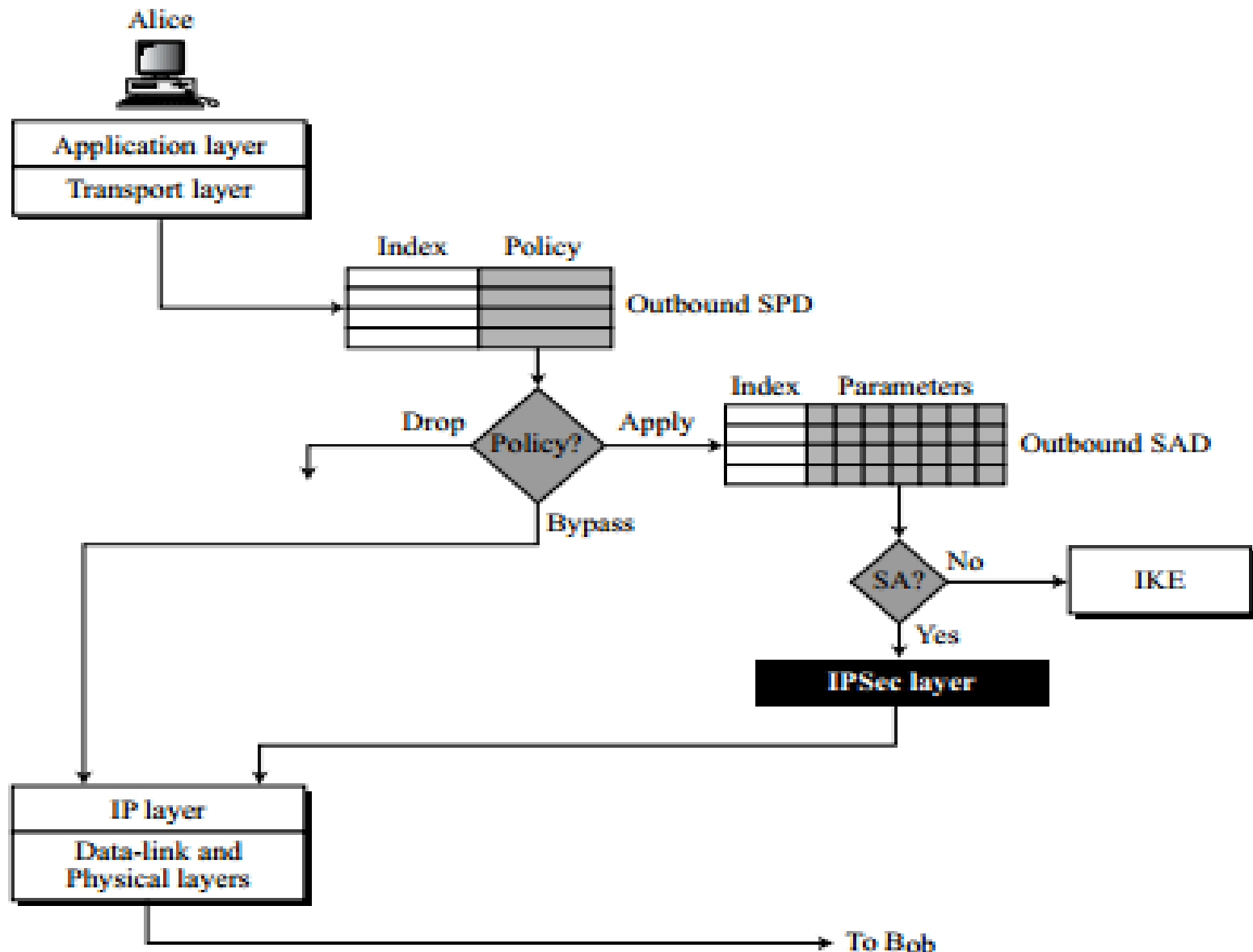
- Each entry in the SPD can be accessed using a six-tuple index: **source address, destination address, name, protocol, source port, and destination port**

Index	Policy
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	

Outbound processing

- When a packet is to be sent out, the outbound SPD is consulted
- The input to the outbound SPD is the six-tuple index
- The output is one of the three following cases:
 - Drop
 - packet defined by the index cannot be sent
 - Bypass
 - there is no policy for the packet with the current policy index
 - Apply
 - the security header is applied

Outbound processing



- If an outbound SA is already established:
 - the triple SA index is returned that selects the corresponding SA from the outbound SAD
 - The AH or ESP header is formed
 - encryption, authentication, or both are applied based on the SA selected
 - The packet is transmitted.
- If an outbound SA is not established yet, the Internet Key Exchange (IKE) protocol is called to create an outbound and inbound SA for this traffic

Inbound processing

- When a packet arrives, the inbound SPD is consulted.
- Each entry in the inbound SPD is accessed using the same six-tuple index
- The output to the inbound SPD can be of following types:
 - Discard
 - Bypass
 - Apply

- If an inbound SA is already established,
 - the triple SA index is returned (selects the corresponding inbound SA from the inbound SAD)
 - Decryption, authentication, or both are applied.
 - If the packet passes the security criteria, the AH or ESP header is discarded and the packet is delivered to the transport layer.
- If an SA is not yet established,
 - the packet must be discarded